Progress Check Presentation

Natural Disaster Severity Prediction

Team [Your Group ID]

Data Mining Final Project
NYCU - Spring 2026
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Project Overview

Goal

• Predict drought severity (0-5 scale) for next 5 weeks
• Using only historical meteorological data

Dataset

• Training: 2,248 regions × 5,480 days = 12.3M rows
• Testing: 2,248 regions × 91 days
• Features: 14 weather variables
- Temperature (tmp, tmp_max, tmp_min, tmp_range)
- Humidity, Precipitation, Wind, Pressure

Current Approach

Model Architecture

• Random Forest Regressor

- 200 trees, max depth 20
- Trained on 1.76M weekly drought scores

Features Used (14)

• Wind: speed, min, max, range

• Temperature: tmp, max, min, range, surface

• Humidity, Precipitation

• Pressure: surface, dew point, wet bulb

Prediction Strategy

• Use last 28 days average of weather data

• Generate 5-week forecast with trend adjustment

Current Results

Performance Metrics

- Validation MAE: 0.8072
- Public Leaderboard MAE: 0.8973
- Current Rank: #6 out of 10 teams

Baseline	Score	Status
Baseline 3	0.8056	■ Need to beat
Baseline 2	0.8623	■ Beat
Baseline 1	0.9117	■ Beat
Our Model	0.8973	Current

Gap to Close: Need ~10% improvement to beat Baseline 3

Data Exploration Findings

Score Distribution (Highly Imbalanced)

Score	Count	Percentage	Severity
0	1,048,333	60%	No drought
1	303,432	17%	Mild
2	186,279	11%	Moderate
3	118,496	7%	Severe
4	69,422	4%	Very Severe
5	31,974	2%	Extreme

Key Patterns

- Scores recorded weekly (every 7 days)
- Most regions have low/no drought (60%)
- Extreme drought is rare (only 2%)

Challenges Encountered

1. Weekly Score Pattern

- Scores only recorded once per 7 days
- Other 6 days are NaN → Need special handling

2. Multi-Week Prediction

- Predicting 5 weeks ahead is harder than 1 week
- Uncertainty increases with time horizon
- No future weather data available

3. Model Performance Gap

- Current score: 0.8973
- Target: < 0.8056 (need ~10% improvement)
- Simple features may not capture complex drought patterns

4. Data Imbalance

- 60% of data has no drought (score 0)
- Model may be biased toward predicting low scores

Next Steps & Improvements

1. Feature Engineering (High Priority)

• Rolling averages (7, 14, 28 days)
• Lag features (previous weeks' data)
• Drought indicators:
- Consecutive dry days
- Heat-dryness index
- Moisture deficit

2. Advanced Models

• XGBoost / LightGBM (gradient boosting)
• Model ensemble (combine multiple models)
• Hyperparameter tuning

3. Better Prediction Strategy

• Trend analysis (is drought worsening?)
• Region-specific patterns
• Temporal feature importance

Timeline:

• Next submission: This week
• Target: Beat Baseline 3 by June 10

Summary

What We've Done

- Downloaded and explored data
- Built baseline Random Forest model
- Successfully submitted to Kaggle (0.8973 MAE)
- Ranked #6 out of 10 teams

What's Working

- Model successfully predicts general drought levels
- Beat 2 out of 3 baselines
- Solid foundation for improvement

What Needs Improvement

- Need better features to capture drought patterns
- Multi-week predictions need refinement
- Must beat Baseline 3 (0.8056)

Confidence Level: Moderate

We have a solid foundation and clear path to improvement.